BIG MOUNTAIN RESORT

Pricing Strategy Recommendations

Executive Presentation | Data Science Team | May 2026

$81.00	$95.87	+$14.87	~$0.88
Current Ticket Price	Model-Supported Price	Pricing Opportunity	New Lift Cost / Ticket

PROBLEM IDENTIFICATION

105	Trails
11	Chair Lifts
Montana	Location

The Business Challenge

New Chairlift Investment

A new chairlift was recently installed, adding $1,540,000 in seasonal operating costs. The business needs to understand how to recover this cost.

Pricing Strategy Unknown

Big Mountain has never had a data-driven pricing model. Current pricing is based on intuition, not competitive intelligence or facility benchmarking.

No Facility-Price Framework

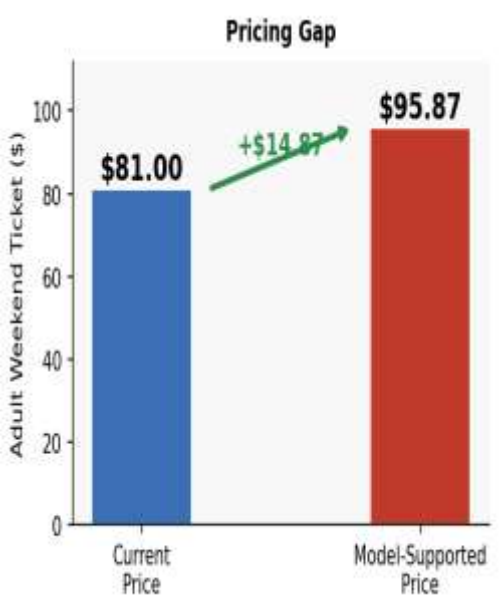
Without knowing which facilities drive ticket price, investment decisions are made without understanding their financial impact.

The Core Finding: Big Mountain Is Underpriced

$14.87

per ticket

pricing opportunity
identified by the model



Big Mountain currently charges $81.00 for an adult weekend ticket.

The model, trained on 277 competitor resorts, supports a price of $95.87.

Even at the model's ~$8 margin of error, a meaningful gap remains.

The new chairlift costs only $0.88 per ticket — the gap is 17x that.

The underpricing predates the new lift — correcting it is the single biggest revenue lever available to the business.

Our Recommendation

RAISE TICKET PRICES from $81 toward $95.87 using a phased approach — $86 in Year 1, $91 in Year 2.
Pursue the vertical drop extension (Scenario 2) as the priority capital investment.

Model Accuracy

$8 MAE

Random Forest model reduces prediction error by 57% vs baseline. CV and test results consistent.

Top Price Drivers

4 Features

Fast quads, vertical drop, snow making, and runs drive ticket price. Big Mountain excels on all four.

Lift Cost Recovery

$0.88/ticket

The new chairlift costs less than $1 per ticket. A $1 price increase alone covers it with surplus.

Modelling Approach & Results

1

Baseline

Predict mean price for all resorts
MAE = $18.80 — our floor to beat

2

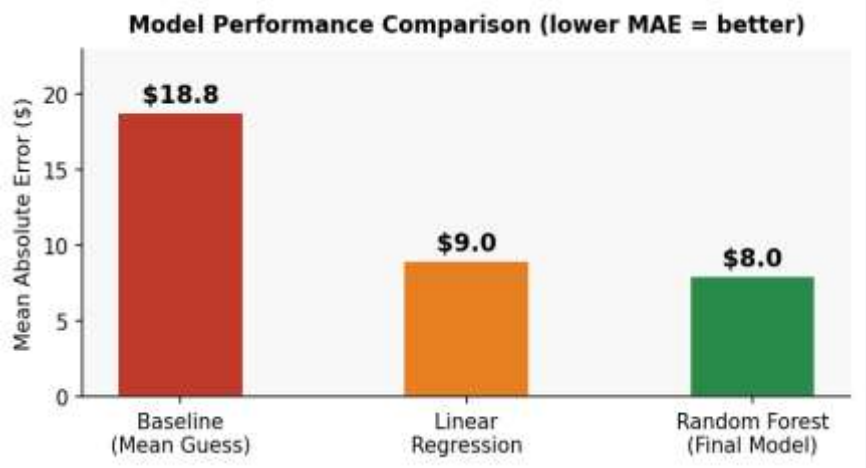
Linear Regression

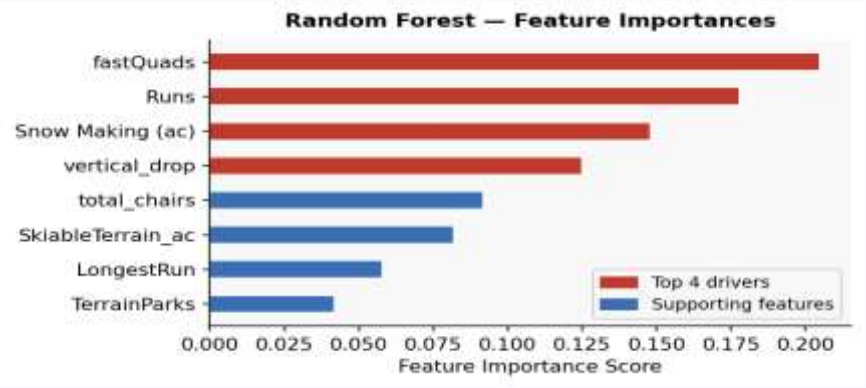
8 best features via GridSearchCV
MAE ≈ $9.00, R^2 = 70%+ on test

3

Random Forest

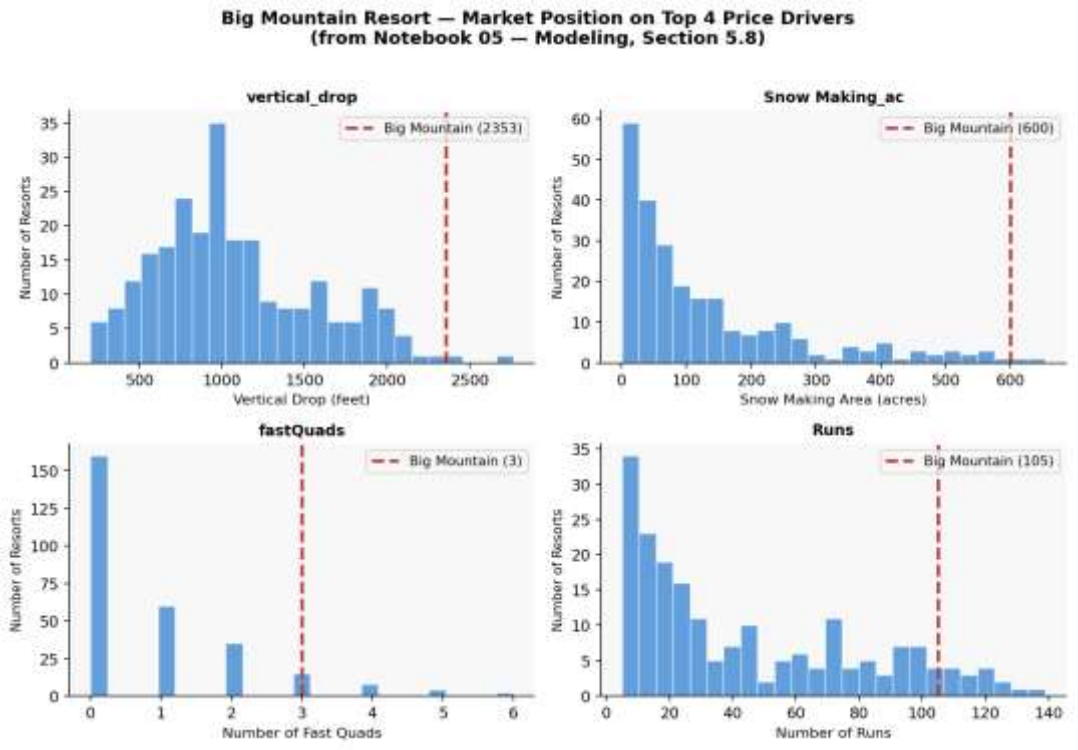
Median imputation, no scaling needed
MAE ≈ $8.00 — selected model





Key: fast quads, vertical drop, snow making & runs drive ticket price

Big Mountain's Market Position



Source: Notebook 05, Section 5.8 — Red dashed line = Big Mountain's value

Vertical Drop
Percentile vs US resorts
81st

Snow Making Area
Percentile vs US resorts
93rd

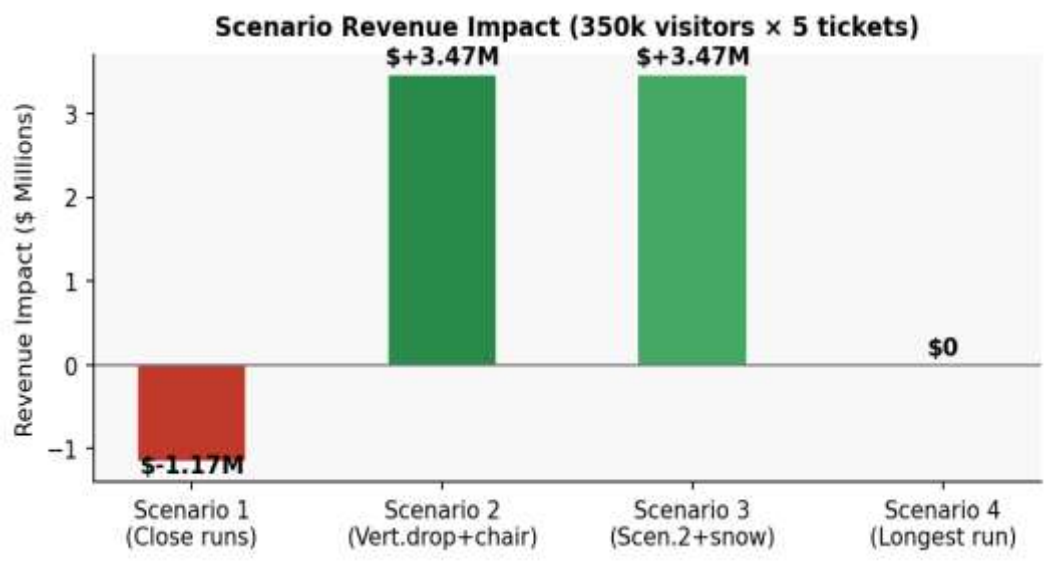
Total Chairs
Percentile vs US resorts
87th

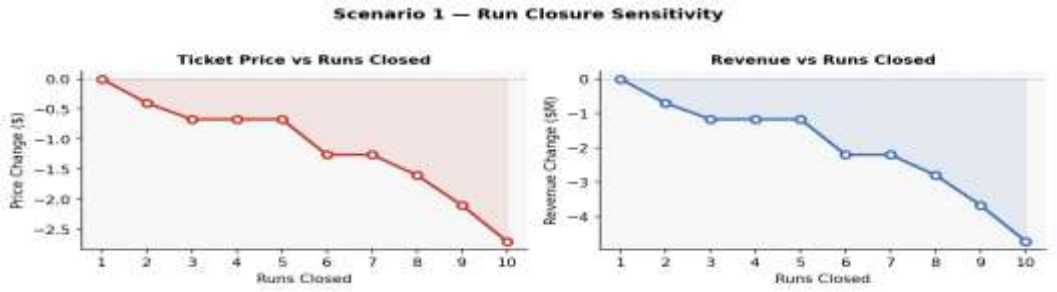
Fast Quads
Percentile vs US resorts
76th

Number of Runs
Percentile vs US resorts
78th

Top quartile on EVERY key price driver — yet charging less than competitors

Scenario Analysis — Investment Decisions





Scenario 1: Closing runs only reduces revenue. Never increases it.

S1 — **Close Runs** — CAUTION

Closing 1 = no effect. Closing 6+ = large revenue drop. Pilot off-peak only.

S2 — **Vert. Drop + Chair** — RECOMMENDED

+$3.47M revenue. Targets the 2 strongest price drivers. Return > 2x lift cost.

S3 — **S2 + Snow Making** — NEUTRAL

Same result as S2. Big Mountain already leads on snow making — 2ac adds nothing.

S4 — **Extend Longest Run** — AVOID

Zero revenue impact. Lowest-ranked feature. Invest elsewhere.

Summary & Next Steps

Key Takeaways

- Big Mountain has elite facilities — top quartile on every key price driver.
- Current price of $81.00 is $14.87 below what the market will support.
- The new chairlift adds only $0.88/ticket — the pricing gap covers it 17×.
- Scenario 2 (vertical drop + chair) projects +$3.47M in additional revenue.
- Run closures must be piloted carefully — 6+ closures cause large revenue drops.

Recommended Actions

1. Raise ticket prices to $86 in Year 1, $91 in Year 2 — phased approach reduces risk.
2. Invest in Scenario 2: vertical drop extension + new chair for +$3.47M revenue.
3. Pilot 1-2 run closures off-peak, measure impact before any permanent decision.
4. Deploy model as self-service tool so analysts can test scenarios independently.